

Sheikh Hefzul Bari, Ph.D.

Graduate School of Engineering || Tohoku University, Japan

Email: shbari.bd@gmail.com || Website: shbari.github.io

CURRENT AFFILIATION

- 2026 - Present **Specially Appointed Assistant Professor (Research)** , Tohoku University
- Developing climate change impact assessment tools using multi-ensemble regional climate projections (d4PDF) to support sustainable watershed management.
- 2026 - Present **Visiting Researcher** , Fukushima University
- General research on watershed hydrology.

EDUCATION

- 2022 - 2026 **Ph.D. in Science and Engineering (Symbiotic Systems Science)** , Fukushima University, Japan
- Dissertation: Monitoring and data analysis of suspended sediment transport from the Abukuma River watershed to the Pacific Ocean.*
- Advisor: Prof. Yoshiyuki Yokoo
- 2012 - 2014 **MSc in Civil & Environmental Engineering** , Shahjalal University of Science & Technology
- 2008 - 2011 **BSc in Civil & Environmental Engineering** , Shahjalal University of Science & Technology

PAST EXPERIENCES

- 2023 - 2026 **Research Assistant** , Fukushima University, Japan
- 2013 - 2022 **Faculty Member** , Leading University, Bangladesh

AWARDS, GRANTS AND FELLOWSHIPS

- 2026 **論文奨励賞 (Paper Encouragement Award)** , Japan Society of Hydrology and Water Resources
- Award recognizing an original and promising research contribution in hydrology and water resources.*
- 2026 **President's Award for Outstanding Research Achievement** , Fukushima University, Japan
- Conferred in recognition of research contributions and publication output during doctoral studies.*
- 2022 - 2026 **Japanese Government (MEXT) Scholarship** , Ministry of Education, Culture, Sports, Science and Technology (MEXT), Japan
- Fully funded competitive Ph.D. fellowship awarded for doctoral studies.*
- 2019 **Leading University Research Grant** , Leading University, Bangladesh
- Research Funding: "Estimating probable maximum precipitation for Bangladesh."*
- 2017 **Leading University Research Grant** , Leading University, Bangladesh
- Research Funding: "Assessing climate change impact in Haor Basin of Bangladesh."*
- 2017 **Best Researcher Award** , Leading University, Bangladesh
- Recognized for outstanding institutional research contributions.*
- 2015 - 2016 **R&D Research Grant** , Ministry of Science and Technology, Bangladesh
- Research Funding: "Development of low-cost desalination device."*
- 2012 - 2014 **National Science and Technology (NST) Fellowship** , Ministry of Science and Technology, Bangladesh
- Graduate fellowship awarded in support of Master's studies.*

RESEARCH & TECHNICAL EXPERTISE

Modeling & Simulation	Hydrological & Hydraulic Modeling Rainfall-Runoff-Inundation (RRI) modeling.
Data & Analytics	Scientific Computing & Geospatial Analysis Python (scientific plotting and data processing), GIS mapping.
Field & Lab Work	Environmental Monitoring & Laboratory Analysis Estuarine water sampling, particle size distribution (PSD) analysis.
Languages	Bengali, English, Japanese & Hindi Native Bengali speaker; proficient in English with experience in scientific writing; basic proficiency in Japanese and Hindi for everyday communication.

INVITED TALKS/LECTURES

May 2025	Bari, SH. and Yokoo, Y., "A reconnaissance study on estuarine suspended sediment dynamics in the Abukuma River, Japan." Invited oral presentation at the Japan Geoscience Union (JpGU) Meeting 2025, Chiba, Japan.
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PUBLICATIONS

(Only Selected Listed)

JOURNALS

1. Ugajin, R., Yokoo, Y., Sayama, T., **Bari, SH.** and Leong, C. (2026). Improving Rainfall-Runoff-Inundation Model for realistic representations of main river with levees. *Hydrological Research Letters*, 20(1), pp.60-66.
2. **Bari, SH.**, Yokoo, Y. and Leong, C. (2026). Temporal dynamics and threshold behaviors of suspended sediment transport in the Abukuma River, Japan. *Hydrological Research Letters*, 20(2), pp.127-134.
3. **Bari, SH.**, Yokoo, Y., Tateno, M. and Leong, C. (2025). Dynamics, particle size distribution and mineral composition of suspended sediment transport in the Abukuma River estuary. *Hydrological Research Letters*, 19(1), pp.51-57.
4. **Bari, SH.**, Yokoo, Y. and Leong, C. (2024). A brief review of recent global trends in suspended sediment estimation studies. *Hydrological Research Letters*, 18(2), pp.51-57.
5. Husna, NEA., **Bari, SH.**, Islam, GT. and Islam, AS. (2024). Groundwater drought assessment in Southwestern Bangladesh. *Int. J. Hydrology Science and Technology*, 17(3), pp.267-277.
6. Husna, NEA., **Bari, SH.**, Islam, GT., Islam, AS. and Hussain, MM. (2024). Modelling drought as a climate change indicator in the southwest coastal region of Bangladesh. *Int. J. Hydrology Science and Technology*, 18(4), pp.351-367.
7. Hussain, MM., Mahmud, I. and **Bari, SH.** (2023). pyHomogeneity: a python package for homogeneity test of time series data. *J. of Open Research Software*, 11(1).
8. **Bari, SH.** and Husna, NEA. (2023). People's perception on climate change effects and adaptation in the Haor basin of Bangladesh. *Int. J. Hydrology Science and Technology*, 15(1), pp.40-53.
9. **Bari, SH.** (2021). Estimating probable maximum precipitation for Bangladesh. *Int. J. Hydrology Science and Technology*, 11(4), pp.415-421.
10. **Bari, SH.**, Hussain, MM. and Husna, NEA. (2017). Rainfall variability and seasonality in northern Bangladesh. *Theoretical and Applied Climatology*, 129(3-4), pp.995-1001.
11. Mahmud, I., **Bari, SH.** and Rahman, MT. (2017). Monthly rainfall forecast of Bangladesh using autoregressive integrated moving average method. *Environmental Engineering Research*, 22(2), pp.162-168.
12. **Bari, SH.**, Hoque, MA., Rahman, MT. and Hussain, MM. (2016). Analysis of seasonal and annual rainfall trends in the northern region of Bangladesh. *Atmospheric Research*, 176-177, pp.148-158.

13. Husna, NEA., **Bari, SH.**, Hussain, MM., Rahman, MT. and Rahman, M. (2016). Ground water level prediction using artificial neural network. *Int. J. Hydrology Science and Technology*, 6(4), pp.371-381.
14. Hussain, MM., **Bari, SH.**, Tarif, ME., Rahman, MT. and Hoque, MA. (2016). Temporal and spatial variation of groundwater level in Mymensingh district, Bangladesh. *Int. J. Hydrology Science and Technology*, 6(2), pp.188-197.
15. **Bari, SH.**, Rahman, MT., Hussain, MM. and Ray, S. (2015). Forecasting Monthly Precipitation in Sylhet City Using ARIMA Model. *Civil and Environmental Research*, 7(1), pp.69-77.

CONFERENCES

1. **Bari, SH.**, Yokoo, Y. and Udo, K. (2026). Multi-temporal suspended sediment dynamics and threshold-controlled sand transport in the Abukuma River estuary, Japan. *Asia Oceania Geosciences Society (AOGS) Annual Meeting 2026* (Poster), Fukuoka, Japan, 2-7 August.
2. **Bari, SH.** and Yokoo, Y. (2025). Characterizing suspended sediment in the Abukuma River estuary: Composition, transport, and estimation. *Japan Society of Hydrology and Water Resources 2025 Research Presentation Conference* (Oral), Akita, Japan, 17-19 September.
3. **Bari, SH.** and Yokoo, Y. (2025). Suspended sediment in the Abukuma River estuary, Japan: Composition, particle size, and hysteresis. *Asia Oceania Geosciences Society (AOGS) Annual Meeting 2025* (Oral), Marina Bay Sands, Singapore, 27 July-1 August.
4. **Bari, SH.**, Tateno, M. and Yokoo, Y. (2024). Suspended sediment transport in the Abukuma River estuary. *The 12th International Symposium on Water Environment Systems* (Poster), Sendai, Japan, 21-22 November.
5. **Bari, SH.**, Tateno, M. and Yokoo, Y. (2024). Estimating sand, silt and clay transport at the Abukuma River estuary for a comprehensive understanding of suspended sediment dynamics from the watershed to the Pacific Ocean. *Japan Geoscience Union (JpGU) Meeting 2024* (Poster), Chiba, Japan, 26-31 May.
6. **Bari, SH.** and Yokoo, Y. (2024). Preliminary field study for understanding estuarine suspended sediment dynamics at the Abukuma River, Japan. *Japan Society of Civil Engineers Annual Meeting 2024* (Oral), Sendai, Japan, 28 February-1 March.

BOOK CHAPTERS

1. Alias, NE., **Bari, SH.**, Paramasivam, R. and Abdullah, FN. (2023). Computations of probable maximum precipitation estimates. In *Handbook of Hydroinformatics- Advances in Machine Learning Techniques*, Elsevier.
2. Hussain, MM., **Bari, SH.**, Mahmud, I. and Siddiquee, MIH. (2021). Application of different artificial neural network for streamflow forecasting. In *Advances in Streamflow Forecasting – from Traditional to Modern Approaches*, Elsevier Limited, Oxford, UK.